

Ten exam questions solved with Exercise 3 and Theorem 4.2

Every question has (a) consistency and (b) the asymptotic distribution. Notation: $\psi_\theta = \partial_\theta m$, $\Psi_n(\theta) = \frac{1}{n} \sum_i \psi_\theta(Z_i)$, $\Psi(\theta) = \mathbb{E}\psi_\theta(Z)$. Exercise 3 is stated for a nondecreasing Ψ_n , so when Ψ_n decreases we apply it to $-\Psi_n$.

When to use Exercise 3(a) and when 3(b)

Both need: (pointwise LLN) $\Psi_n(\theta) \xrightarrow{\mathbb{P}} \Psi(\theta)$ for each fixed θ , and (sign change) $\Psi(\theta_0 - \varepsilon) < 0 < \Psi(\theta_0 + \varepsilon)$. They differ in what they ask of Ψ_n :

3(a): Ψ_n is *continuous* and has *exactly one* zero $\hat{\theta}_n$ (on an event with probability $\rightarrow 1$ is enough).

Typical: a smooth, strictly monotone score whose limits at the ends of Θ have opposite signs, e.g. $+\infty \rightarrow -\infty$ or $+1 \rightarrow -1$. Uniqueness comes from strict monotonicity, existence from the limits.

3(b): Ψ_n is *monotone*, and $\Psi_n(\hat{\theta}_n) = o_{\mathbb{P}}(1)$ (an extra “near-zero” check).

Typical: (1) Ψ_n is a *staircase*, so there is no exact zero (quantile, median); (2) Ψ_n has *flat parts*, so the zero is not unique (Huber); (3) the zero may *not exist* on some samples (GLMs, mixture at the boundary).

Quick test. Is Ψ_n continuous? *No* $\Rightarrow$ 3(b). *Yes*: can you show, from the limits of Ψ_n at the ends of Θ , that it has exactly one zero? *Yes* $\Rightarrow$ 3(a); *no* (flat parts, or the zero may not exist) $\Rightarrow$ 3(b). If Ψ_n is monotone, 3(b) always works; 3(a) just saves the near-zero check.

1. Weibull shape

$Z_1, \dots, Z_n$ iid with density $f(z | k) = kz^{k-1}e^{-z^k}$, $z > 0$, $k \in (0, \infty)$. (a) Show that the MLE $\hat{k}_n$ is consistent. (b) Find its asymptotic distribution.

(a) Consistency.

Use Exercise 3(a) for $-\Psi_n$. Ψ_n is continuous and strictly decreasing, from $+\infty$ to $-\infty$, so it has exactly one zero (with probability $\rightarrow 1$). (No explicit $\hat{k}_n$: k sits inside Z_i^k .)

Using $\frac{d}{dk} z^k = z^k \log z$,

$$\ell = \log k + (k-1) \log z - z^k, \quad \psi_k(z) = \frac{1}{k} + \log z - z^k \log z, \quad \partial_k \psi_k(z) = -\frac{1}{k^2} - z^k (\log z)^2 < 0.$$

For expectations: $W = Z^{k_0} \sim \text{Exp}(1)$, since $\mathbb{P}(Z^{k_0} > w) = e^{-w}$, and $\mathbb{E} \log W = \Gamma'(1) = -\gamma$, $\mathbb{E}[W \log W] = \Gamma'(2) = 1 - \gamma$.

- (i) *Continuous with exactly one zero.* Ψ_n is continuous and strictly decreasing, since $\partial_k \psi < 0$. As $k \downarrow 0$, $\frac{1}{k} \rightarrow \infty$, so $\Psi_n \rightarrow +\infty$. As $k \rightarrow \infty$, $Z_i^k \log Z_i \rightarrow \infty$ for any $Z_i > 1$, so $\Psi_n \rightarrow -\infty$; this fails only if all $Z_i \leq 1$, which has probability $(1 - e^{-1})^n \rightarrow 0$. Hence there is exactly one zero with probability $\rightarrow 1$.
- (ii) *Pointwise convergence.* $|\psi_k| \leq \frac{1}{k} + |\log Z| + Z^k |\log Z|$, which is integrable (all moments of W and $\log W$ are finite). By the WLLN, $\Psi_n(k) \xrightarrow{\mathbb{P}} \Psi(k)$.
- (iii) *Sign change.* $\Psi(k_0) = \frac{1}{k_0} (1 + \mathbb{E} \log W - \mathbb{E}[W \log W]) = \frac{1}{k_0} (1 - \gamma - 1 + \gamma) = 0$. Since $\psi_k(z)$ is strictly decreasing in k for every z , Ψ is strictly decreasing, so $\Psi(k_0 - \varepsilon) > 0 > \Psi(k_0 + \varepsilon)$.

By Exercise 3(a), $\hat{k}_n \xrightarrow{\mathbb{P}} k_0$.

(b) Asymptotic distribution. We check Theorem 4.2.

C1 Shown in (a).

C2 ℓ is C^∞ in k , k_0 is interior to $(0, \infty)$, and the support $(0, \infty)$ does not depend on k .

C3 $\Psi_n(k_0)$ is an iid average with mean $\Psi(k_0) = 0$ and finite variance (moments of W), so $\sqrt{n}\Psi_n(k_0) \xrightarrow{d} N(0, I)$ by Theorem 1.6.

C4 On $[k_0/2, 2k_0]$ we have $z^k \leq z^{k_0/2} + z^{2k_0}$, so $|\partial_k \psi| \leq \frac{4}{k_0^2} + (z^{k_0/2} + z^{2k_0})(\log z)^2 =: d(z)$ with $\mathbb{E}d < \infty$. By Lemma 4.1 the Hessian converges uniformly.

It is an MLE, so $\Sigma = -H_0 = I$ (Lemma 4.2). Using $\mathbb{E}[W(\log W)^2] = \Gamma''(2) = \frac{\pi^2}{6} - 1 + (1 - \gamma)^2$,

$$I = -\mathbb{E}\partial_k \psi_{k_0} = \frac{1 + \mathbb{E}[W(\log W)^2]}{k_0^2} = \frac{\pi^2/6 + (1 - \gamma)^2}{k_0^2} = \frac{1.8237}{k_0^2}.$$

$$\sqrt{n}(\hat{k}_n - k_0) \xrightarrow{d} N(0, k_0^2/(\pi^2/6 + (1 - \gamma)^2)) \approx N(0, 0.548 k_0^2).$$

2. Logistic location

$Z_1, \dots, Z_n$ iid with density $f(z | \theta) = \frac{e^{-(z-\theta)}}{(1 + e^{-(z-\theta)})^2}$, $\theta \in \mathbb{R}$; write $\sigma(u) = 1/(1 + e^{-u})$. (a) Show that the MLE is consistent. (b) Find its asymptotic distribution.

(a) Consistency.

Use Exercise 3(a) for $-\Psi_n$. Ψ_n is continuous and strictly decreasing, from $+1$ to -1 , so it has exactly one zero for every sample.

With $u = z - \theta$, $\ell = -u - 2\log(1 + e^{-u})$, so $\partial_u \ell = -1 + 2(1 - \sigma(u)) = 1 - 2\sigma(u)$. Since $\partial_\theta = -\partial_u$ and $\sigma' = \sigma(1 - \sigma)$,

$$\psi_\theta(z) = 2\sigma(z - \theta) - 1, \quad \partial_\theta \psi_\theta(z) = -2\sigma(z - \theta)(1 - \sigma(z - \theta)) \in [-\tfrac{1}{2}, 0).$$

Since the logistic cdf is σ itself, $V = \sigma(Z - \theta_0) \sim \text{Unif}(0, 1)$.

(i) *Continuous with exactly one zero.* Ψ_n is continuous and strictly decreasing, from 1 (as $\theta \rightarrow -\infty$) to -1 (as $\theta \rightarrow \infty$), so it has exactly one zero.

(ii) *Pointwise convergence.* $|\psi| \leq 1$, so the WLLN applies.

(iii) *Sign change.* $\Psi(\theta_0) = \mathbb{E}[2V - 1] = 0$, and Ψ is strictly decreasing because each $\psi_\theta(z)$ is. Hence $\Psi(\theta_0 - \varepsilon) > 0 > \Psi(\theta_0 + \varepsilon)$.

By Exercise 3(a), $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

(b) Asymptotic distribution.

C1 Shown in (a).

C2 C^∞ in θ ; $\Theta = \mathbb{R}$ is open; the support $\mathbb{R}$ does not depend on θ .

C3 The mean is 0, and $\Sigma = \mathbb{E}(2V - 1)^2 = 4 \text{Var } V = \frac{4}{12} = \frac{1}{3}$. Theorem 1.6 applies.

C4 $|\partial_\theta \psi| \leq \frac{1}{2}$ is a constant envelope (Lemma 4.1). $H_0 = -2\mathbb{E}[V(1 - V)] = -2(\frac{1}{2} - \frac{1}{3}) = -\frac{1}{3}$.

$\Sigma = -H_0 = \frac{1}{3}$, as Lemma 4.2 predicts. Variance $= \Sigma/H_0^2 = 3$.

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, 3).$$

3. Exponential claim-severity regression

(X_i, Z_i) iid, $|X| \leq K$, $\mathbb{P}(X > 0) > 0$, $\mathbb{P}(X < 0) > 0$. Given X , $Z \sim \text{Exp}$ with rate $e^{\beta_0 X}$. (a) Show that the MLE $\hat{\beta}_n$ is consistent. (b) Find its asymptotic distribution.

(a) Consistency.

Use Exercise 3(a) for $-\Psi_n$. Ψ_n is continuous and strictly decreasing, and its limits are $+\infty$ and $-\infty$ as soon as both signs of X occur, so it has exactly one zero with probability $\rightarrow 1$.

$$\ell = \beta X - Z e^{\beta X}, \quad \psi_\beta = X(1 - Z e^{\beta X}), \quad \partial_\beta \psi_\beta = -Z X^2 e^{\beta X} \leq 0.$$

Given X , $W = Z e^{\beta_0 X} \sim \text{Exp}(1)$, so $\mathbb{E}[Z \mid X] = e^{-\beta_0 X}$ and, by the tower property, $\Psi(\beta) = \mathbb{E}[X(1 - e^{(\beta - \beta_0)X})]$.

- (i) *Continuous with exactly one zero.* Ψ_n is continuous and strictly decreasing once some $X_i \neq 0$. As $\beta \rightarrow \infty$, a term with $X_i > 0$ gives $-Z_i X_i e^{\beta X_i} \rightarrow -\infty$; as $\beta \rightarrow -\infty$, a term with $X_i < 0$ gives $Z_i |X_i| e^{\beta X_i} \rightarrow +\infty$. Both signs of X occur with probability $\geq 1 - \mathbb{P}(X \leq 0)^n - \mathbb{P}(X \geq 0)^n \rightarrow 1$, so there is exactly one zero with probability $\rightarrow 1$.
- (ii) *Pointwise convergence.* $|\psi_\beta| \leq K(1 + Z e^{|\beta|K})$ and $\mathbb{E}Z \leq e^{|\beta_0|K}$, so the WLLN applies.
- (iii) *Sign change.* $\Psi(\beta_0) = 0$. For $\beta > \beta_0$, $X(1 - e^{(\beta - \beta_0)X}) < 0$ whenever $X \neq 0$ (both factors have opposite signs). So $\Psi(\beta_0 + \varepsilon) < 0$, and likewise $\Psi(\beta_0 - \varepsilon) > 0$.

By Exercise 3(a), $\hat{\beta}_n \xrightarrow{\mathbb{P}} \beta_0$.

(b) Asymptotic distribution.

C1 Shown in (a).

C2 Smooth in β ; $\Theta = \mathbb{R}$; the support $(0, \infty)$ does not depend on β .

C3 $\mathbb{E}\psi_{\beta_0} = \mathbb{E}[X(1 - \mathbb{E}[W \mid X])] = 0$, and $\Sigma = \mathbb{E}[X^2 \text{Var}(W \mid X)] = \mathbb{E}X^2$.

C4 On $[\beta_0 - 1, \beta_0 + 1]$, $|\partial_\beta \psi| \leq Z K^2 e^{(|\beta_0|+1)K} =: d$ with $\mathbb{E}d < \infty$. $H_0 = -\mathbb{E}[X^2 e^{\beta_0 X} \mathbb{E}[Z \mid X]] = -\mathbb{E}X^2$.

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(0, 1/\mathbb{E}X^2).$$

4. Poisson regression

(X_i, Y_i) iid, $|X| \leq K$, $\mathbb{P}(X \neq 0) > 0$, $Y \mid X \sim \text{Poisson}(e^{\beta_0 X})$. (a) Show that the MLE $\hat{\beta}_n$ is consistent. (b) Find its asymptotic distribution.

(a) Consistency.

Use Exercise 3(b) for $-\Psi_n$. Ψ_n is monotone, but the zero need not exist: if all $Y_i = 0$ and all $X_i > 0$, then $\Psi_n < 0$ everywhere and there is no MLE. Existence must come from the posts, which is the near-zero check (iv).

$$\ell = Y\beta X - e^{\beta X} - \log Y!, \quad \psi_\beta = (Y - e^{\beta X})X, \quad \partial_\beta \psi_\beta = -e^{\beta X} X^2 \leq 0,$$

and by the tower property $\Psi(\beta) = \mathbb{E}[(e^{\beta_0 X} - e^{\beta X})X]$.

- (i) *Monotone.* $\partial_\beta \psi \leq 0$, so Ψ_n is nonincreasing.
- (ii) *Pointwise convergence.* $|\psi_\beta| \leq K(Y + e^{|\beta|K})$ and $\mathbb{E}Y \leq e^{|\beta_0|K}$, so the WLLN applies.

- (iii) *Sign change.* $\Psi(\beta_0) = 0$. For $\beta > \beta_0$, $(e^{\beta_0 X} - e^{\beta X})X < 0$ whenever $X \neq 0$. Hence $\Psi(\beta_0 + \varepsilon) < 0 < \Psi(\beta_0 - \varepsilon)$.
- (iv) *Near-zero.* By (ii)–(iii), with probability $\rightarrow 1$, $\Psi_n(\beta_0 - 1) > 0 > \Psi_n(\beta_0 + 1)$. Ψ_n is continuous, so a root exists (IVT) and $\Psi_n(\hat{\beta}_n) = 0$. Hence $\Psi_n(\hat{\beta}_n) = o_{\mathbb{P}}(1)$.

By Exercise 3(b), $\hat{\beta}_n \xrightarrow{\mathbb{P}} \beta_0$.

(b) Asymptotic distribution.

C1 Shown in (a).

C2 Smooth; $\Theta = \mathbb{R}$; the support $\{0, 1, \dots\}$ does not depend on β .

C3 $\mathbb{E}\psi_{\beta_0} = \mathbb{E}[X(\mathbb{E}[Y | X] - e^{\beta_0 X})] = 0$, and $\Sigma = \mathbb{E}[X^2 \text{Var}(Y | X)] = \mathbb{E}[e^{\beta_0 X} X^2]$.

C4 On $[\beta_0 - 1, \beta_0 + 1]$, $|\partial_{\beta}\psi| \leq K^2 e^{(|\beta_0|+1)K}$, a constant. $H_0 = -\mathbb{E}[e^{\beta_0 X} X^2]$.

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(0, 1/\mathbb{E}[e^{\beta_0 X} X^2]).$$

5. Logistic regression

(X_i, Y_i) iid, $\mathbb{E}X^2 < \infty$, $\mathbb{P}(X \neq 0) > 0$, $Y | X \sim \text{Bernoulli}(\sigma(\beta_0 X))$. (a) Show that the MLE is consistent. (b) Find its asymptotic distribution.

(a) Consistency.

Use Exercise 3(b) for $-\Psi_n$. Ψ_n is monotone, but the zero need not exist: under perfect separation ($Y_i = 1$ iff $X_i > 0$) every term is positive, so $\Psi_n > 0$ everywhere.

$$\ell = Y\beta X - \log(1 + e^{\beta X}), \quad \psi_{\beta} = (Y - \sigma(\beta X))X, \quad \partial_{\beta}\psi_{\beta} = -\sigma(1 - \sigma)(\beta X) X^2 \leq 0.$$

Write $\sigma_0 = \sigma(\beta_0 X)$; then $\Psi(\beta) = \mathbb{E}[(\sigma_0 - \sigma(\beta X))X]$.

- (i) *Monotone.* $\partial_{\beta}\psi \leq 0$.
- (ii) *Pointwise convergence.* $|\psi| \leq |X|$, which is integrable; WLLN.
- (iii) *Sign change.* $\Psi(\beta_0) = 0$. σ is strictly increasing, so for $\beta > \beta_0$, $(\sigma_0 - \sigma(\beta X))X < 0$ whenever $X \neq 0$. Hence $\Psi(\beta_0 + \varepsilon) < 0 < \Psi(\beta_0 - \varepsilon)$.
- (iv) *Near-zero.* As in Q4: with probability $\rightarrow 1$ the values at $\beta_0 \pm 1$ have opposite signs, so by continuity a root exists and $\Psi_n(\hat{\beta}_n) = 0$.

By Exercise 3(b), $\hat{\beta}_n \xrightarrow{\mathbb{P}} \beta_0$.

(b) Asymptotic distribution.

C1 Shown in (a).

C2 Smooth; $\Theta = \mathbb{R}$; $Y \in \{0, 1\}$ for every β .

C3 Mean 0; $\Sigma = \mathbb{E}[X^2 \text{Var}(Y | X)] = \mathbb{E}[\sigma_0(1 - \sigma_0)X^2] \leq \frac{1}{4}\mathbb{E}X^2$.

C4 $|\partial_{\beta}\psi| \leq \frac{1}{4}X^2 =: d$ for all β , with $\mathbb{E}d < \infty$. $H_0 = -\mathbb{E}[\sigma_0(1 - \sigma_0)X^2]$.

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(0, 1/\mathbb{E}[\sigma_0(1 - \sigma_0)X^2]). \text{ (E.g. } X = \pm 1, \beta_0 = 0: \text{ variance 4.)}$$

6. Mixture weight

f_0, f_1 known densities with $f_0 \neq f_1$ on a set of positive measure. Z_i iid with density $g_\pi = \pi f_1 + (1 - \pi)f_0$, $\pi \in (0, 1)$, true value $\pi_0 \in (0, 1)$. (a) Show that the MLE $\hat{\pi}_n$ is consistent. (b) Find its asymptotic distribution.

(a) Consistency.

Use Exercise 3(b) for $-\Psi_n$. Ψ_n is monotone, but the zero need not lie in $(0, 1)$: if $f_1(Z_i) > f_0(Z_i)$ for all i , then $\Psi_n > 0$ on $(0, 1)$ and the MLE is at the boundary $\pi = 1$.

$$\psi_\pi = \frac{f_1 - f_0}{g_\pi}, \quad \partial_\pi \psi_\pi = -\frac{(f_1 - f_0)^2}{g_\pi^2} = -\psi_\pi^2 \leq 0.$$

Since $g_\pi \geq \min(\pi, 1 - \pi)(f_0 + f_1) \geq \min(\pi, 1 - \pi)|f_1 - f_0|$, we have $|\psi_\pi| \leq 1/\min(\pi, 1 - \pi)$. Take $\varepsilon < \min(\pi_0, 1 - \pi_0)$ so that both posts lie in $(0, 1)$.

- (i) *Monotone.* $\partial_\pi \psi = -\psi^2 \leq 0$.
- (ii) *Pointwise convergence.* ψ_π is bounded for fixed $\pi \in (0, 1)$; WLLN.
- (iii) *Sign change.* $\Psi(\pi_0) = \int \frac{f_1 - f_0}{g_{\pi_0}} g_{\pi_0} dz = \int f_1 - \int f_0 = 0$. For $\pi_1 < \pi_2$,

$$\psi_{\pi_1} - \psi_{\pi_2} = \frac{(f_1 - f_0)^2(\pi_2 - \pi_1)}{g_{\pi_1} g_{\pi_2}} \geq 0,$$

strictly where $f_1 \neq f_0$. So Ψ is strictly decreasing and $\Psi(\pi_0 - \varepsilon) > 0 > \Psi(\pi_0 + \varepsilon)$.

- (iv) *Near-zero.* With probability $\rightarrow 1$ the posts have opposite signs; Ψ_n is continuous, so a root exists in $(0, 1)$ and $\Psi_n(\hat{\pi}_n) = 0$.

By Exercise 3(b), $\hat{\pi}_n \xrightarrow{\mathbb{P}} \pi_0$.

(b) Asymptotic distribution.

- C1** Shown in (a).
- C2** Smooth in π ; π_0 interior; the support $\{f_0 > 0\} \cup \{f_1 > 0\}$ is the same for all $\pi \in (0, 1)$.
- C3** Mean 0 by (iii); $\Sigma = \mathbb{E}\psi_{\pi_0}^2 = \int \frac{(f_1 - f_0)^2}{g_{\pi_0}} dz < \infty$ (since ψ is bounded).
- C4** With $\delta = \frac{1}{2} \min(\pi_0, 1 - \pi_0)$, on $[\pi_0 - \delta, \pi_0 + \delta]$: $|\partial_\pi \psi| = \psi^2 \leq 1/\delta^2$, a constant. $H_0 = -\mathbb{E}\psi_{\pi_0}^2 = -\Sigma$.

$\sqrt{n}(\hat{\pi}_n - \pi_0) \xrightarrow{d} N\left(0, \left[\int \frac{(f_1 - f_0)^2}{\pi_0 f_1 + (1 - \pi_0) f_0} dz\right]^{-1}\right)$. (For $f_0 = \text{Exp}(1)$, $f_1 = \text{Exp}(0.2)$, $\pi_0 = 0.2$, numerically: $I \approx 2.413$, variance ≈ 0.414 .)

7. Log-cosh location (not an MLE)

Z_i iid with density symmetric about θ_0 . $\hat{\theta}_n$ maximises $-\frac{1}{n} \sum_i \log \cosh(Z_i - \theta)$. (a) Show consistency. (b) Find the asymptotic distribution.

(a) Consistency.

Use Exercise 3(a) for $-\Psi_n$. Ψ_n is continuous and strictly decreasing, from $+1$ to -1 , so it has exactly one zero for every sample.

$$m = -\log \cosh(z - \theta), \quad \psi_\theta = \tanh(z - \theta), \quad \partial_\theta \psi_\theta = -\operatorname{sech}^2(z - \theta) \in [-1, 0].$$

Write $U = Z - \theta_0$.

- (i) *Continuous with exactly one zero.* Ψ_n is continuous and strictly decreasing from 1 to -1 .
- (ii) *Pointwise convergence.* $|\tanh| \leq 1$; WLLN.
- (iii) *Sign change.* $\tanh$ is odd and $U \stackrel{d}{=} -U$, so $\Psi(\theta_0) = \mathbb{E} \tanh U = 0$. Ψ is strictly decreasing because each $\psi_\theta(z)$ is.

By Exercise 3(a), $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

(b) Asymptotic distribution.

C1 Shown in (a).

C2 C^∞ in θ ; $\Theta = \mathbb{R}$.

C3 Mean 0; $\Sigma = \mathbb{E} \tanh^2 U \leq 1$.

C4 $|\partial_\theta \psi| \leq 1$, a constant. $H_0 = -\mathbb{E} \operatorname{sech}^2 U < 0$.

This is not an MLE, so we keep the sandwich Σ/H_0^2 . For $U \sim N(0, 1)$: $\mathbb{E} \tanh^2 U = 0.394$, and by $\tanh^2 + \operatorname{sech}^2 = 1$, $\mathbb{E} \operatorname{sech}^2 U = 0.606$.

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \mathbb{E} \tanh^2 U / (\mathbb{E} \operatorname{sech}^2 U)^2); \text{ for } N(\theta_0, 1): 0.394/0.606^2 \approx 1.075.$$

8. Huber location (non-smooth)

Z_i iid with continuous density f symmetric about θ_0 , $f > 0$ near θ_0 . Fix $k > 0$, $\psi_k(u) = \max(-k, \min(k, u))$; $\hat{\theta}_n$ solves $\sum_i \psi_k(Z_i - \theta) = 0$. (a) Show consistency. (b) Find the asymptotic distribution.

(a) Consistency.

Use Exercise 3(b) for $-\Psi_n$. Ψ_n is continuous and monotone but can be *flat*: on any stretch where no residual $Z_i - \theta$ lies in $(-k, k)$, every term is $\pm k$. So the zero set can be an interval and “exactly one zero” fails.

$\psi_\theta(z) = \psi_k(z - \theta)$ is 1-Lipschitz in θ , with derivative $-\mathbf{1}_{(|z-\theta|<k)}$ except at the kinks $\theta = z \pm k$. Write $U = Z - \theta_0$. Since $\psi_k(a) - \psi_k(b) = \int_b^a \mathbf{1}_{(|s|<k)} ds$, Fubini gives

$$\Psi(\theta) - \Psi(\theta') = \int_\theta^{\theta'} \mathbb{P}(|Z - t| < k) dt \quad (\theta < \theta'), \quad \text{so} \quad \Psi'(\theta) = -\mathbb{P}(|Z - \theta| < k).$$

- (i) *Monotone.* ψ_k is nondecreasing and $z - \theta$ decreases in θ , so Ψ_n is nonincreasing.
- (ii) *Pointwise convergence.* $|\psi_k| \leq k$; WLLN.
- (iii) *Sign change.* $\Psi(\theta_0) = \mathbb{E} \psi_k(U) = 0$, since ψ_k is odd and U is symmetric. By the formula above, $\Psi(\theta_0 - \varepsilon) = \int_{\theta_0 - \varepsilon}^{\theta_0} \mathbb{P}(|Z - t| < k) dt > 0$ (because $f > 0$ near θ_0); likewise $\Psi(\theta_0 + \varepsilon) < 0$.
- (iv) *Near-zero.* Ψ_n is continuous with limits $\pm k$, so a zero always exists and $\Psi_n(\hat{\theta}_n) = 0$.

By Exercise 3(b), $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

(b) Asymptotic distribution. m is not C^2 (kinks), so C2/C4 fail; we use the Exercise 7 route.

- N1** Take the curvature from the smooth population curve: $H_0 = \Psi'(\theta_0) = -\mathbb{P}(|U| < k)$, continuous in θ .
- N2** $\mathbb{G}_n \psi_{\theta_0} = n^{-1/2} \sum \psi_k(U_i) \xrightarrow{d} N(0, \Sigma)$ with $\Sigma = \mathbb{E} \psi_k(U)^2 \leq k^2$ (Theorem 1.6).
- N3** Donsker. $|\psi_\theta - \psi_{\theta'}| \leq |\theta - \theta'|$, so on $[\theta_0 - 1, \theta_0 + 1]$ a grid with spacing ε gives brackets $[\psi_{\theta_j} \pm \frac{\varepsilon}{2}]$ of size ε . Then $N_{[]} \leq 2/\varepsilon + 1$ and $J_{[]} < \infty$, so by Theorem 1.19 the class is Donsker. Since $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$, $\mathbb{G}_n \psi_{\hat{\theta}_n} - \mathbb{G}_n \psi_{\theta_0} = o_{\mathbb{P}}(1)$.
- N4** $0 = \sqrt{n} \Psi_n(\hat{\theta}_n) = \mathbb{G}_n \psi_{\hat{\theta}_n} + \sqrt{n} \Psi(\hat{\theta}_n) = \mathbb{G}_n \psi_{\theta_0} + o_{\mathbb{P}}(1) + \Psi'(\bar{\theta}_n) \sqrt{n}(\hat{\theta}_n - \theta_0)$, and $\Psi'(\bar{\theta}_n) \xrightarrow{\mathbb{P}} H_0$. By Slutsky, $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \Sigma/H_0^2)$.
- For $U \sim N(0, 1)$, $k = 1$: $H_0 = -0.6827$ and $\Sigma = \mathbb{E} \min(U^2, 1) = (0.6827 - 2\varphi(1)) + 2(1 - \Phi(1)) = 0.1988 + 0.3173 = 0.5161$.

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \mathbb{E} \psi_k(U)^2 / \mathbb{P}(|U| < k)^2); \text{ for } N(\theta_0, 1), k = 1: 0.5161/0.6827^2 \approx 1.107.$$

9. Sample τ -quantile (non-smooth)

$\tau \in (0, 1)$. Z_i iid with cdf F , density f continuous and > 0 near $\theta_0 = F^{-1}(\tau)$. $\hat{\theta}_n$ minimises $\sum_i \rho_\tau(Z_i - \theta)$, $\rho_\tau(u) = u(\tau - \mathbf{1}_{(u < 0)})$. (a) Show consistency. (b) Find the asymptotic distribution.

(a) Consistency.

Use Exercise 3(b). $\Psi_n = F_n - \tau$ is a *staircase*: it jumps by $\frac{1}{n}$ at each data point, so it generally never equals 0 (unless $n\tau$ is an integer). 3(a) is impossible; 3(b) only needs $\Psi_n(\hat{\theta}_n) \approx 0$.

For $\theta < z$, $\rho_\tau = \tau(z - \theta)$ has θ -derivative $-\tau$; for $\theta > z$, $\rho_\tau = (\tau - 1)(z - \theta)$ has derivative $1 - \tau$. So

$$\psi_\theta(z) = \mathbf{1}_{(z \leq \theta)} - \tau, \quad \Psi_n(\theta) = F_n(\theta) - \tau, \quad \Psi(\theta) = F(\theta) - \tau,$$

already nondecreasing (we minimise), so no sign flip is needed.

- (i) *Monotone.* F_n is nondecreasing.
- (ii) *Pointwise convergence.* $F_n(\theta)$ is an average of iid Bernoulli($F(\theta)$); WLLN.
- (iii) *Sign change.* $\Psi(\theta_0) = 0$ by the definition of θ_0 ; $f > 0$ near θ_0 , so $F(\theta_0 - \varepsilon) < \tau < F(\theta_0 + \varepsilon)$.
- (iv) *Near-zero.* At the minimum, left derivative $\leq 0 \leq$ right derivative: $F_n(\hat{\theta}_n -) \leq \tau \leq F_n(\hat{\theta}_n)$. With no ties the jump is $\frac{1}{n}$, so $0 \leq \Psi_n(\hat{\theta}_n) \leq \frac{1}{n}$.

By Exercise 3(b), $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

(b) Asymptotic distribution. $\Psi'_n = 0$ between jumps, so C4 fails; we use the Exercise 7 route.

- N1** $H_0 = \Psi'(\theta_0) = f(\theta_0) > 0$.
- N2** $\mathbb{G}_n \psi_{\theta_0} = n^{-1/2} \sum (\mathbf{1}_{(Z_i \leq \theta_0)} - \tau) \xrightarrow{d} N(0, \tau(1 - \tau))$.
- N3** Donsker. Choose $-\infty = t_0 < \dots < t_K = \infty$ with $F(t_j) - F(t_{j-1}) \leq \varepsilon^2$, so $K \leq 1/\varepsilon^2 + 1$. For $\theta \in [t_{j-1}, t_j]$, $\mathbf{1}_{(z \leq t_{j-1})} \leq \mathbf{1}_{(z \leq \theta)} \leq \mathbf{1}_{(z \leq t_j)}$, a bracket of L^2 -size $\leq \varepsilon$. $J_{[]} < \infty$, so by Theorem 1.19 $\mathbb{G}_n \psi_{\hat{\theta}_n} - \mathbb{G}_n \psi_{\theta_0} = o_{\mathbb{P}}(1)$.
- N4** $\sqrt{n} \Psi_n(\hat{\theta}_n) \in [0, n^{-1/2}]$, and $\sqrt{n} \Psi_n(\hat{\theta}_n) = \mathbb{G}_n \psi_{\theta_0} + o_{\mathbb{P}}(1) + f(\bar{\theta}_n) \sqrt{n}(\hat{\theta}_n - \theta_0)$ with $f(\bar{\theta}_n) \xrightarrow{\mathbb{P}} f(\theta_0)$. Slutsky.

$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \tau(1-\tau)/f(\theta_0)^2)$. (E.g. Exp(1), $\tau = 0.9$: $\theta_0 = \log 10$, $f(\theta_0) = 0.1$, variance 9.)

10. Laplace location MLE (non-smooth)

Z_i iid with density $f(z | \theta) = \frac{1}{2}e^{-|z-\theta|}$. (a) Show that the MLE is consistent. (b) Find its asymptotic distribution.

(a) Consistency.

Use Exercise 3(b) for $-\Psi_n$. Ψ_n is a *staircase* (jumps of $\frac{2}{n}$), with no exact zero when n is odd.

$\ell = -\log 2 - |z - \theta|$, so $\psi_\theta(z) = \text{sign}(z - \theta)$, and the MLE is the sample median. $\Psi_n(\theta) = 1 - 2F_n(\theta)$ a.e., and $\Psi(\theta) = 1 - 2F(\theta)$. We verify the conditions for $-\Psi_n = 2F_n - 1$.

- (i) *Monotone.* $2F_n - 1$ is nondecreasing.
- (ii) *Pointwise convergence.* Bounded terms; WLLN.
- (iii) *Sign change.* $F(\theta_0) = \frac{1}{2}$ by symmetry and F is strictly increasing, so $2F(\theta_0 - \varepsilon) - 1 < 0 < 2F(\theta_0 + \varepsilon) - 1$.
- (iv) *Near-zero.* The right and left derivatives of $\frac{1}{n} \sum |Z_i - \theta|$ are $2F_n(\theta) - 1$ and $2F_n(\theta-) - 1$. At the minimum they straddle 0, and with no ties the jump is $\frac{2}{n}$, so $0 \leq -\Psi_n(\hat{\theta}_n) \leq \frac{2}{n}$.

By Exercise 3(b), $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

(b) Asymptotic distribution. ℓ has a kink and $\ell'' = 0$ a.e., so Theorem 4.3 / C4 fail; we use the Exercise 7 route.

N1 $H_0 = \Psi'(\theta_0) = -2f(\theta_0 | \theta_0) = -2 \cdot \frac{1}{2} = -1$.

N2 $\text{sign}(Z_i - \theta_0) = \pm 1$ with probability $\frac{1}{2}$ each: mean 0, $\Sigma = 1$; Theorem 1.6.

N3 $\text{sign}(z - \theta) = 1 - 2 \cdot \mathbf{1}_{(z \leq \theta)}$: the indicator brackets of Q9 (with F -steps $\leq \varepsilon^2/4$) give Donsker, so $\mathbb{G}_n \psi_{\hat{\theta}_n} - \mathbb{G}_n \psi_{\theta_0} = o_{\mathbb{P}}(1)$.

N4 $\sqrt{n}\Psi_n(\hat{\theta}_n) = O(n^{-1/2}) = \mathbb{G}_n \psi_{\theta_0} + o_{\mathbb{P}}(1) + \Psi'(\bar{\theta}_n)\sqrt{n}(\hat{\theta}_n - \theta_0)$, with $\Psi'(\bar{\theta}_n) \xrightarrow{\mathbb{P}} -1$. Slutsky.

Variance $= \Sigma/H_0^2 = 1$. The Fisher information is $I = \mathbb{E} \text{sign}^2 = 1$, so this equals I^{-1} even though Lemma 4.2 cannot be used ($\mathbb{E}\ell'' = 0$).

$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, 1) = N(0, I^{-1})$.